

Ismail Hossain

PhD Candidate in Computer Science

📧 ismail102.github.io

✉ ihossain@miners.utep.edu

+1 915-503-9122

🔗 Google Scholar

🔍 RESEARCH INTERESTS

Agentic AI Security · LLM Safety & Robustness · Adversarial Machine Learning · Privacy-Preserving ML (Federated Learning)

🎓 EDUCATION

University of Texas at El Paso (UTEP) El Paso, TX
PhD in Computer Science (Expected Graduation - December, 2026) Aug. 2024 – Present
Advisor: Dr. Sajedul Talukder · Lab: SUPREME Lab (Security and Privacy-Enhanced Machine Learning)
Dissertation focus: LLM safety evaluation, agentic AI security, and privacy-preserving machine learning.
University of Texas at El Paso (UTEP) El Paso, TX
MS in Computer Science May 2025
Chittagong University of Engineering & Technology (CUET) Chittagong, Bangladesh
Bachelor of Science in Computer Science and Engineering Mar. 2014 – Nov. 2018

👤 TEACHING EXPERIENCE

Graduate Teaching Assistant UTEP & SIUC
Department of Computer Science Aug. 2023 – Present

- Courses taught as instructor of record / lab instructor: Design & Analysis of Algorithms, Learning from Data, Ethical Hacking, Computer Security, Computer Networks, Computer Programming (Python/C++).
- Delivered lectures, designed lab exercises, and created assessments for undergraduate and graduate students; class sizes ranging from 20 to 60 students.
- Provided one-on-one mentoring in algorithms, cybersecurity principles, and machine learning — consistently receiving positive student feedback.
- Developed supplementary course materials, including lecture slides, coding assignments, and rubrics aligned to course learning outcomes.

Teaching Interests (courses ready to teach from Day 1):

- Machine Learning / Deep Learning
- Artificial Intelligence
- LLMs & Generative AI (new course)
- Algorithms & Data Structures
- Computer Security
- Privacy & Security in Computing

🧰 RESEARCH & INDUSTRY EXPERIENCE

🏢 Graduate Research Associate El Paso, TX
Security and Privacy Enhanced Machine Learning Lab. Aug. 2022 – Present

- Lead researcher on LLM safety, jailbreak evaluation, and agentic AI security — resulting in submissions to ACM CCS'26, NeurIPS'26, and AAAI'26.
- Designed and released **optimus-jbscorer** (PyPI), an open-source LLM safety scoring library with the Optimus two-dimensional metric $J(S, H)$.
- Fine-tuned transformer LLMs (LoRA, PEFT, quantization) on multi-GPU clusters using PyTorch with FSDP and DDP distributed training.
- Built end-to-end ML pipelines: data curation, instruction tuning, RAG integration, and large-scale evaluation across 114K+ prompt datasets.
- Designed multi-agent LLM systems (LangChain, AutoGen) for safety-aware, tool-augmented applications.
- Constructed and managed LAMDA — a large-scale longitudinal Android malware benchmark accepted at ICLR'26.
- Mentored 8+ graduate and undergraduate researchers; 6 mentees co-authored published papers.

- Full-stack developer for the Data Information Management Studio (telecom domain); built the Golden Parameter Management System for centralized configuration validation.
- Designed scalable web applications (Angular + Spring Boot), RESTful APIs, and SQL/NoSQL database layers supporting millions of data records.
- Delivered 800+ commits, 100+ pull requests, and 300+ code reviews across global distributed teams following Agile/CI-CD practices.
- Recognized as Top Contributor; selected for a technology-transfer business trip to Samsung Digital City, Suwon, South Korea.

PUBLICATIONS

Peer-Reviewed Conference Papers

2026

- Hossain, I. et al. “The Art of the Jailbreak: Formulating Jailbreak Attacks for LLM Security Beyond Binary Scoring.”  [ACM CCS 2026](#) - Under Review.

Builds a 114K cybersecurity-grounded compositional jailbreak dataset with instruction fine-tuned generators (perplexity 24–39 vs. 40–140 for baselines); proposes Optimus, a training-free two-dimensional metric $J(S, H)$ that exposes a stealth-optimal regime ($S^* \approx 0.57, H^* \approx 0.43$) that binary ASR entirely collapses.

- Hossain, I., Puppala, S., Lu, Z., Talukder, S., & Jiang, N. “Benchmarking Security Risk Detection and Verification in Open Agentic Skill Ecosystems.” [NeurIPS 2026](#) - Under Review.

Presents SkillVetBench, combining LLM-based semantic vetting over skill specifications with sandboxed execution to verify threats; shows static baselines miss up to 89% of malicious skills, with runtime attacks concentrated in `exec`, `write_file`, `install_skill`, and `spawn` primitives.

- Hossain, I., Puppala, S., Alam, M.J., & Talukder, S. “AI-in-the-Loop: Privacy Preserving Real-Time Scam Detection and Conversational Scambaiting by Leveraging LLMs and Federated Learning.”  [PoPETs 2026](#), Issue 1 - Acceptance Rate: 20%.

A privacy-preserving AI-in-the-loop framework that proactively detects and disrupts scam conversations in real time using instruction-tuned LLMs with a safety-aware utility function; federated learning enables continual updates without raw data sharing, achieving perplexity as low as 22.3 and engagement ≈ 0.80 .

- Hossain, I., et al. “When Safety Geometry Collapses: Fine-Tuning Vulnerabilities in Agentic Guard Models.”  [AAAI 2026 Summer Symposium Series](#).

Shows that guard models (LlamaGuard, WildGuard, Granite Guardian) fine-tuned on entirely benign data lose all safety alignment via destruction of latent safety geometry; Granite Guardian’s refusal rate drops from 85% to 0% with CKA falling to zero, explained by the specialization hypothesis.

- Hossain, I., et al. “Agent-Fence: Mapping Security Vulnerabilities Across Deep Research Agents.”  [AAAI 2026 Summer Symposium Series](#).

Defines 14 trust-boundary attack classes across planning, memory, retrieval, tool use, and delegation; evaluates 8 agent archetypes under persistent multi-turn interaction, finding mean security break rates (MSBR) ranging from 0.29 (LangGraph) to 0.51 (AutoGPT), with Denial-of-Wallet and Authorization Confusion as the highest-risk classes.

- Hossain, I., et al. “Semantic Intent Fragmentation: A Single-Shot Compositional Attack on Multi-Agent AI Pipelines.”  [AAAI 2026 Summer Symposium Series](#).

Introduces SIF, an attack where a single legitimate request causes an orchestrator to decompose tasks into individually benign subtasks that jointly violate security policy; a GPT-20B orchestrator produces policy-violating plans in 71% of 14 enterprise scenarios with 0% false positives from the cross-model compliance judge.

- Haque, M.A., Hossain, I., Kamol, M.M., Alam, M.J., Amalapuram, S.K., Talukder, S., & Rahman, M.S. “LAMDA: A Longitudinal Android Malware Benchmark for Concept Drift Analysis.”  [ICLR 2026](#) - Acceptance Rate: 28%.

The largest temporally diverse Android malware benchmark: 12 years (2013–2025), 1M+ samples, 1,380 malware families, and 150,000 singleton samples; empirically quantifies ML model performance degradation over time due to concept drift and analyzes feature stability across years.

2025

- Puppala, S., Hossain, I., Alam, M.J., Ahad, T., & Talukder, S. “LLM-Guided Dynamic-UMAP for Personalized Federated Graph Learning.” [🔗 CIKM 2025](#).

Proposes VGM² (Variational Gaussian Mixture Manifold), a geometry-centric personalized federated learning framework that learns client-specific parametric UMAP embeddings to handle label skew and non-stationarity across heterogeneous clients.

- Alam, M.J., Hossain, I., Puppala, S., & Talukder, S. “Comprehensive Privacy Risk Assessment in Social Networks Using User Attributes, Social Graphs, and Text Analysis.” [🔗 ACM HyperText 2025](#) - Acceptance Rate: 33%.

Presents CPRS (Comprehensive Privacy Risk Scoring), a framework that quantifies privacy risk by integrating user attribute profiles, social graph structures, and user-generated text to produce a unified, interpretable privacy risk score.

- Hossain, I., Puppala, S., Alam, M.J., & Talukder, S. “Toward Empathetic AI: Neural-Symbolic LLMs for Emotionally Aligned Conversations.” [ASONAM 2025](#).

Explores neural-symbolic integration in LLMs to produce emotionally aligned, context-sensitive conversational responses; evaluates empathy quality and alignment across diverse user interaction scenarios.

2024

- Hossain, I., Puppala, S., Alam, M.J., Talukder, S., & Talukder, Z. “EMOVIS: A Visual Approach to Tracking Emotional Sentiment Dynamics in Social Network Commentaries.” [🔗 ICWSM 2024](#) - Acceptance Rate: 20%.

An interactive visualization system that tracks how emotional sentiment evolves across social media comment threads over time, combining NLP-based sentiment extraction with dynamic graph layouts to surface community-level emotional trends.

- Hossain, I., Puppala, S., Alam, M.J., & Talukder, S. “SocialRec: User Activity Based Post Weighted Dynamic Personalized Post Recommendation System in Social Media.” [🔗 ASONAM 2024](#) - Acceptance Rate: 24%.

Analyzes user history, engagement, and persona to generate personalized post recommendations on platforms such as Facebook, Twitter, and Reddit; evaluates ranking quality via Hit Rate (HR) and Normalized Discounted Cumulative Gain (NDCG).

- Hossain, I., Puppala, S., Alam, M.J., & Talukder, S. “EVOLVE: Predicting User Evolution and Network Dynamics in Social Media Using Fine-Tuned GPT-like Model.” [🔗 ASONAM 2024](#) - Acceptance Rate: 24%. Fine-tunes E-GPT (Evolution-GPT), a decoder-only model, to predict future stages of a user’s lifecycle on social media; forecasts future connections and shifts in user activities by modeling demographic attributes and network dynamics over time.

2023

- Hossain, I., Puppala, S., Alam, M.J., & Talukder, S. “Monitoring Dynamics of Emotional Sentiment in Social Network Commentaries.” [ASONAM 2023](#).

Tracks temporal evolution of emotional sentiment across social network comment threads; provides a systematic approach to understanding how community sentiment shifts in response to events and interactions over time.

2022

- Hossain, I., Puppala, S., & Talukder, S. “Collaborative Differentially Private Federated Learning Framework for the Prediction of Diabetic Retinopathy.” [IEEE ICAIC 2022](#).

Introduces a differentially private federated learning framework for medical image classification; enables collaborative model training across healthcare institutions without sharing sensitive patient data.

- Puppala, S., Hossain, I., & Talukder, S. “Towards Federated Learning Based Contraband Detection Within Airport Baggage X-Rays.” [IEEE ICMLANT 2022](#).

Applies federated learning to contraband detection in X-ray imagery across distributed airport nodes, enabling privacy-preserving model updates without sharing raw security screening data.

Journal Articles

- Hossain, I., Puppala, S., Alam, M.J., & Talukder, S. “A Visual Approach to Tracking Emotional Sentiment Dynamics in Social Network Commentaries.” Springer Nature (journal extension of ICWSM’24).
Extended journal version presenting the full EMOVIS system with additional datasets, user studies, and longitudinal analysis of sentiment evolution across multiple social media platforms.
- Hossain, I., Puppala, S., Alam, M.J., & Talukder, S. “Natural Language Processing for Predictive Maintenance in Nuclear Power Plant: Digital Twin Context.” Transactions of the American Nuclear Society.
Applies NLP to operational log analysis in nuclear power plant digital twins, enabling predictive maintenance by detecting anomaly patterns in natural-language maintenance records and sensor-linked text streams.
- Hossain, I., Puppala, S., Alam, M.J., & Talukder, S. “Enhancing the Cybersecurity of Advanced Nuclear Power Plants through Generative AI and Large Language Models.” Transactions of the American Nuclear Society.
Investigates the application of generative AI and LLMs to detect, explain, and respond to cybersecurity threats in advanced nuclear power plant control systems, emphasizing human-AI collaboration in high-stakes environments.

Book Chapters

- “Blockchain-Integrated Secure Framework for Enhanced E-Government Services.” In Building Cybersecurity Applications with Blockchain and Smart Contracts, Springer.
Proposes a blockchain-based framework for securing e-government service delivery, ensuring data integrity and access auditability across public administration workflows.
- “Differentially Private Federated Learning in Medical Context: Phenomenal Classification of Diabetic Retinopathy.” In Distributed Machine Learning and Optimization, Springer.
Demonstrates how differential privacy combined with federated learning achieves clinically accurate diabetic retinopathy classification while preserving patient confidentiality across distributed hospital nodes.
- “Introduction of Medical Imaging Modalities.” In Data Driven Approaches on Medical Imaging, Springer.
Surveys the landscape of medical imaging modalities — MRI, CT, ultrasound, PET — and their properties as inputs to data-driven machine learning pipelines.
- “Active Learning on Medical Image.” In Data Driven Approaches on Medical Imaging, Springer.
Reviews active learning strategies for reducing annotation burden in medical image datasets, with emphasis on uncertainty sampling and query-by-committee methods for clinical imaging tasks.
- “AutoML Systems for Medical Imaging.” In Data-Driven Approaches on Medical Imaging, Springer.
Examines automated machine learning pipelines tailored to medical imaging, covering neural architecture search, hyperparameter optimization, and domain-specific transfer learning strategies.

MENTORSHIP & ADVISING

Mentored 8+ graduate and undergraduate researchers at UTEP and SIUC on end-to-end research: problem formulation, literature review, experimental design, and conference-quality paper writing. Six mentees have co-authored accepted publications.

Tanzim Ahad 

University of Texas at El Paso

PhD, Year 1

- LLM fine-tuning, post-training alignment, and robustness analysis.
- Co-authored ACM CCS’26 submission on jailbreak evaluation.

Jannatul Ferdous 

University of Texas at El Paso

MS, Year 1

- Federated learning system design and ML pipeline architecture.
- Ongoing thesis research in privacy-preserving machine learning.

Mahedi Hasan 

Green University Bangladesh

BS, Graduated

- Computer vision, YOLO v7–v9, and multimodal Transformer architectures.

Lorenz Lat 

California State University, Stanislaus

BS, Graduated

- Literature review strategies, ML fundamentals, and deep learning workflows.

Francisco Roman	
University of Texas at El Paso	BS, Graduated
User study design, data preprocessing, and fine-tuning dataset preparation.	
Sebastian Lucero-Chavez  & Alejandro Granados 	
University of Texas at El Paso	BS, Current
Data annotation, open-source LLM instruction fine-tuning, and evaluation benchmarking.	

AWARDS & HONORS

3rd Place, Business Idea Competition  Spring 2026 · Miners Pitch Competition, UTEP
 Pitched an AI-driven security startup concept to a panel of industry judges and investors at the University of Texas at El Paso's competitive entrepreneurship showcase. Competed against 47 participants from multiple disciplines and placed 3rd, demonstrating the real-world potential of research in LLM safety and agentic AI.

Student Travel Award (\$1,300) 2024 · ICWSM Conference, Buffalo, NY
 Awarded a competitive travel grant to present research at the International AAAI Conference on Web and Social Media (ICWSM) in Buffalo, NY. The award recognized the quality and impact of the accepted paper on emotional sentiment dynamics in social networks and supported full conference participation.

Distinguished Student Service Award  2024 · Southern Illinois University Carbondale
 Recognized by Southern Illinois University Carbondale for exceptional contributions to the academic community beyond research responsibilities. This honor reflects consistent engagement in mentoring junior students, supporting departmental initiatives, and fostering a collaborative research environment within the Computer Science program.

16th in USA / 4th in Illinois — U.S. Dept. of Energy CyberForce Competition  2022 · National Cybersecurity Competition
 Led a graduate team to a top-20 national finish in the U.S. Department of Energy's CyberForce Competition, a rigorous hands-on cybersecurity defense challenge involving real-world infrastructure scenarios. The team placed 4th in Illinois, competing against universities nationwide and demonstrating applied expertise in network security, system hardening, and incident response.

Outstanding Collaborator Award  2021 · Samsung R&D Institute Bangladesh
 Recognized by Samsung R&D Institute Bangladesh for exemplary cross-functional collaboration on the Data Information Management Studio project. This award is given to engineers who demonstrate outstanding teamwork, communication, and technical contribution across distributed global development teams, reflecting consistent delivery and a positive impact on team culture.

Business Travel Award (\$2,000) 2020 · Samsung Digital City, Suwon, South Korea
 Selected as one of a small group of engineers from Samsung R&D Institute Bangladesh to participate in a technology-transfer trip to Samsung's global headquarters at Digital City in Suwon, South Korea. This competitive award recognized top-performing contributors and provided direct exposure to Samsung's flagship engineering teams and enterprise-scale software practices.

Silver Award, Code Review Contest 2020 · Samsung Electronics
 Earned Silver in Samsung Electronics' internal code review competition, which evaluates engineers on code quality, documentation clarity, adherence to best practices, and constructive peer feedback. The award reflects a strong commitment to software craftsmanship and engineering excellence within one of the world's largest technology organizations.

Advanced Level Certificate 2019 · Samsung Global Competitive Programming Contest

Achieved Advanced Level certification in Samsung’s internal global competitive programming contest, a company-wide algorithmic challenge that tests problem-solving speed, correctness, and efficiency. This level is attained by a select group of participants and reflects strong foundations in data structures, algorithm design, and computational thinking.

ACM ICPC Participant2018 · Dhaka Regional

Qualified for and competed in the ACM International Collegiate Programming Contest (ICPC) at the Dhaka Regional level, one of the most prestigious competitive programming contests for university students worldwide. Participation required passing institutional selection rounds and demonstrated proficiency in algorithms, mathematics, and team-based problem solving under time pressure.

TECHNICAL SKILLS

Languages	Python (primary), C, C++, Java, TypeScript
ML Frameworks	PyTorch (primary), FSDP, DDP, Scikit-learn, HuggingFace Transformers
LLM & Agents	Instruction Tuning, PEFT/LoRA, RAG, PPO/DPO/GRPO, LangChain
AI/ML Areas	Deep Learning, Adversarial ML, Safety Evaluation, Red-Teaming, Federated Learning, Multimodal Learning, Distributed Training
Web & Backend	Angular, Spring Boot, Node.js, Flask, RESTful APIs
Databases	MySQL, PostgreSQL, MongoDB
Infrastructure	Git/GitHub, Docker, CI/CD, Large-scale Data Pipelines, ETL
Open Source	Author: <code>optimus-jbscorer</code> on PyPI; active GitHub contributor
Competitive CP	1,100+ problems solved (LeetCode, Codeforces, UVA, HackerRank)

MEDIA COVERAGE

- SocialRec: Revolutionizing Social Media Personalization with Advanced AI Devdiscourse News, July 2024 · Coverage of: “SocialRec: User Activity Based Post Weighted Dynamic Personalized Post Recommendation System in Social Media”, ASONAM 2024.
- SocFedGPT: Federated GPT-based Adaptive Content Filtering for Social Networks The Moonlight Review, 2024 · Coverage of: “SocFedGPT: Federated GPT-based Adaptive Content Filtering System Leveraging User Interactions in Social Networks”, ASONAM 2024.
- Safety Geometry Collapse: Fine-Tuning Vulnerabilities in Agentic Guard Models Alabia AI Insights, 2026 · Coverage of: “When Safety Geometry Collapses: Fine-Tuning Vulnerabilities in Agentic Guard Models”, AAAI 2026 Summer Symposium Series.

SERVICE

- Reviewer: (20+ peer reviews) - AAAI, ICWSM, ASONAM, WWW, CogSci.
- Open-Source Maintainer: `optimus-jbscorer` (PyPI); actively responds to issues and maintains documentation.
- Competitive Programming Community: Mentored junior students in algorithmic problem solving; participated in ACM ICPC Dhaka Regional 2018.
- CyberForce Competition: Led team to 16th nationally and 4th in Illinois in U.S. Dept. of Energy cybersecurity competition (2022).

REFERENCES

Dr. Sajedul Talukder (PhD Advisor) · SUPREME Lab, UTEP · stalukder@utep.edu